Chapter - 11 : Underground Cables

Book - VK Mehta

- Advantages of underground cables
  - → Less liable through storm and lightning
  - → Low maintanance cost
  - → Less chances of fault
  - → Smaller voltage drop
  - → Better general appearance

- Drawbacks :
  - → Greater installation cost
  - → Introduce insulation problem at high voltages.

- Where use :
  - → Thickly populated areas (reason of safety)
  - → Around plants and ~~substaitied~~ substations.

→ For distribution of electric power in congested areas at comparatively low or moderate voltages.

Cable Requirements :

→ Conductors used in cables should be turned standard Cu or Al of high conductivity.

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→ Conductor size should be such that cable carries that desired load current without overheating and causing voltage drop within permissiable limit.

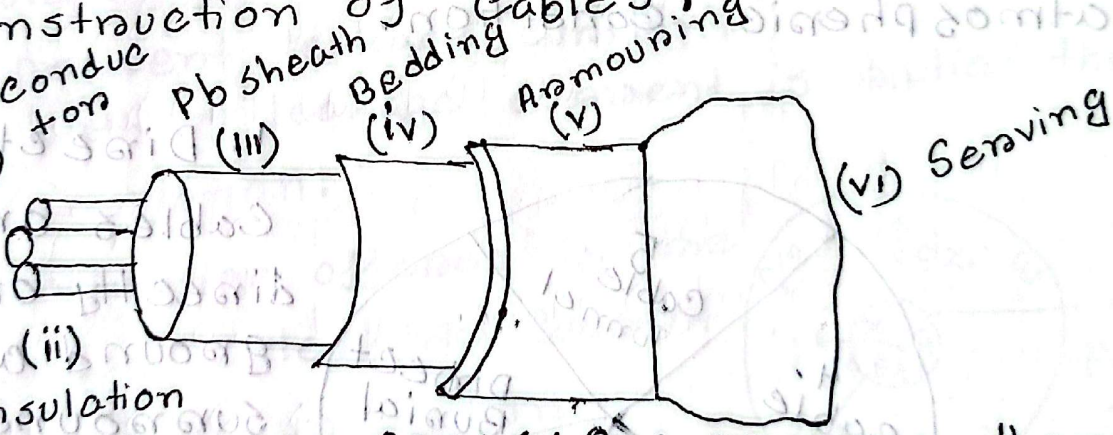
→ Must have proper thickness of insulation to give high degree of safety and reliability

→ Must have suitable mechanical protection

→ Must have chemical and physical stability.

underground cable
- Sylhet, Dhaka
- Monipur, Tejgao .... Dhanmondi > DPDC
- Rajshahi, CTG, Rangpur, khulna

Construction of Cables:



i. Conductors or cores: One or more than one core (conductor) depending on service

ii. Insulation: Suitable thickness of insulator needed. (Material: Impregnated paper, Varnished cambric or rubber mineral compound.

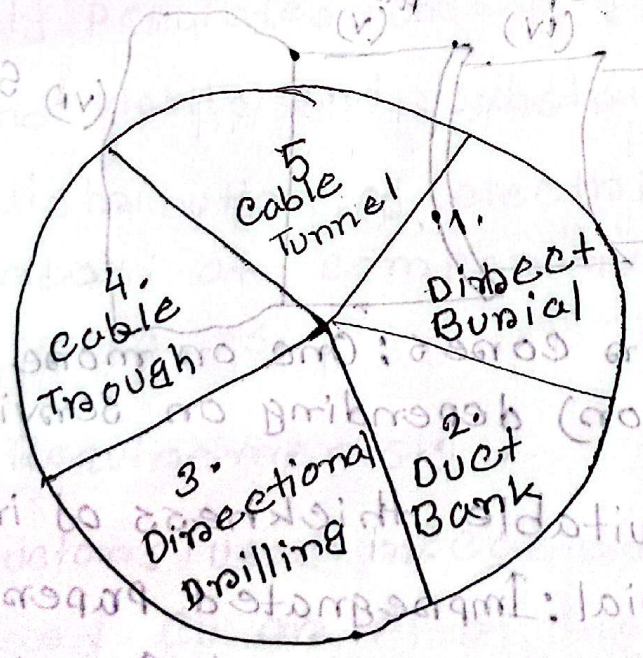
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iii. Metalic Sheath : To protect the cable from moisture, gases or other damaging liquids (acids or alkaines) in the soil and atmosphere (Material : Pb or Al)

iv. Bedding : Fibrous material like Jute, to protect the metalic shield against corrosion and from mechanical injury.

v. Armouring : To protect the cable from mechanical injury while laying it and during the course of handling.

vi. Serving : To protect the armouring from atmospheric condition.



Out of Syllabus

1. Direct Burial : Cables are laid directly on the ground and surrounded with clean back fill sand.

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2. Duct Bank:

Art - 11.1, 11.2 Introduction

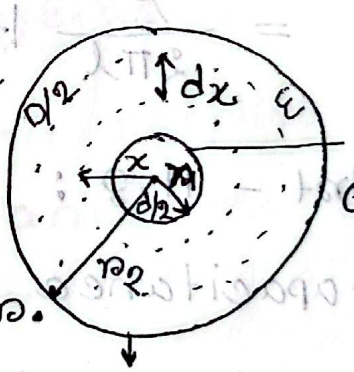
- Survey
- Communication
- Insulation
- Duct
- Sand layer

Self study - Art. 11.3, 11.4, 11.5, 11.6 (Reading)

Lecture - 08
28.07.26

Insulation Resistance of Single Core Cable

- To prevent leakage current
- The path of leakage current is radial through the insulation.
- The opposition offered by the insulation to leakage current is known as insulation resistance of the conductor.



Here, conductor radius = r_1
Internal sheath radius = r_2

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Length of the cable = l

Resistivity of insulation = ρ

Very small layer of insulation thickness at a distance x = dx.

Leakage current flow through length = dx

Leakage current flow through area A = 2πxl

Insulation resistance of considered layer

$$R' = \rho \frac{dx}{2\pi x l} \quad \text{As, } \rho = R/A$$

Insulation resistance of whole cable = R

$$R = \int \rho \frac{dx}{2\pi x l}$$

$$= \frac{\rho}{2\pi l} \int_{r_1}^{r_2} \frac{dx}{x}$$

$$= \frac{\rho}{2\pi l} \ln\left(\frac{r_2}{r_1}\right)$$

Ans - 11.8 :

Capacitance of a single core cable

Single core cable conductor diameter = d

Inner sheath diameter = D

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charges per metre axial length of the cable = Q

Permitivity of the insulation material between core and lead sheath = ε

Consider a cylinder of radius x m.

Axial length = 1m.

Surface Area of the cylinder = $2\pi x \cdot 1$

$= 2\pi x \ m^2$

Electric flux density at any part 'P' on the considered cylinder = Dx

$$D_x = \frac{Q}{\text{Area}} = \frac{Q}{2\pi x} \ C/m^2$$

Electric intensity at point P

$$E_x = \frac{D_x}{\varepsilon} = \frac{Q}{2\pi x \varepsilon} = \frac{Q}{2\pi x \varepsilon_0 \varepsilon_r} \ V/m$$

Work done in moving a unit positive change from point P through a distance dx in the direction of electric field.

$$W_x = E_x dx$$

Work done in moving a unit positive change from conductor to sheath, which is the

Potential diff. V between conducton and sheath, $V = \int_{d/2}^{D/2} E_x dx = \int_{d/2}^{D/2} \frac{Q}{2\pi x \varepsilon_0 \varepsilon_r} dx$

$$= \frac{Q}{2\pi \varepsilon_0 \varepsilon_x} \ln\left(\frac{D}{d}\right)$$

Capacitance of Cable

$$C = \frac{Q}{V} = \frac{2\pi \varepsilon_0 \varepsilon_r}{\ln(D/d)} \text{ F/m}$$

$$= \frac{\varepsilon_r \times 10^{-9}}{41.4 \log(D/d)} \text{ F/m}$$

If conducton length 'l',

$$C = \frac{\varepsilon_r \cdot l}{41.4 \log(D/d)} \times 10^{-9} \text{ F}$$

Presentation

24.08.26

11.9: Dielectric stress in a single core cable

Dielectric stress: $E_x = g$ = potential gradient/ change in voltage with respect to distance.

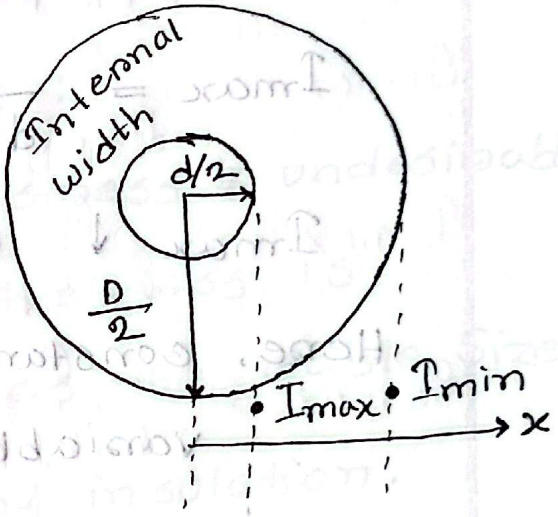
As, $g = E_x = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x}$ ... ①

$$V = \frac{Q}{2\pi\varepsilon_0\varepsilon_r}\ln\left(\frac{D}{d}\right)$$

$$Q = \frac{V(2\pi\varepsilon_0\varepsilon_r)}{\ln(D/d)} \quad \text{... ②}$$

$$g = \frac{V\,2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)(2\pi\varepsilon_0\varepsilon_r)x}$$

$$= \frac{V}{x\ln(D/d)} \text{ V/m}$$



So, $g \propto \frac{1}{x}$

Dielectric stress,

$g_{max} \longrightarrow x_{min}$; $x = \frac{d}{2}$ (surface of conductor)

$g_{min} \longrightarrow x_{max}$; $x = \frac{D}{2}$

$$g_{max} = \frac{2V}{d\ln(D/d)} \text{ V/m}$$

$$g_{min} = \frac{2V}{D\ln(D/d)} \text{ V/m}$$

sat sun mon tue wed thu fri Date :

$$\frac{g_{max}}{g_{min}} = \frac{D}{d}$$

Max stress is important conditions in the design of a cable.

Art. 11.10 :

Most economical conductor size in a cable

$$I_{max} = \frac{2V}{d \log_e (D/d)} \text{ V/m}$$

$I_{max} \downarrow$ $d \log_e (D/d) \uparrow$

Here, constant D, V

variable d

$$\frac{d}{dd}\left[d \log_e \left(\frac{D}{d}\right)\right] = 0$$

$$d = \frac{D}{2.718}$$

$$I_{max} = \frac{2V}{dx}$$

Most economical $(I_{max})_{min}$ value

* Dielectric strength of the insulation should be more than max stress.

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* If conductor's operating voltage is 'V'

$I_{max} \longrightarrow$ 5 kV/mm

Inductor's ~~dista~~ dielectric stress $\geq$ 5 kV/mm

If, Insulator's dielectric stress < 5 kV/mm, breakdown will occur and it will work as conductor.

11.11 : Grading of Cables :

Non linear dielectric stress is undesirable

1. Insulation of greater thickness is required which increases the cable size.

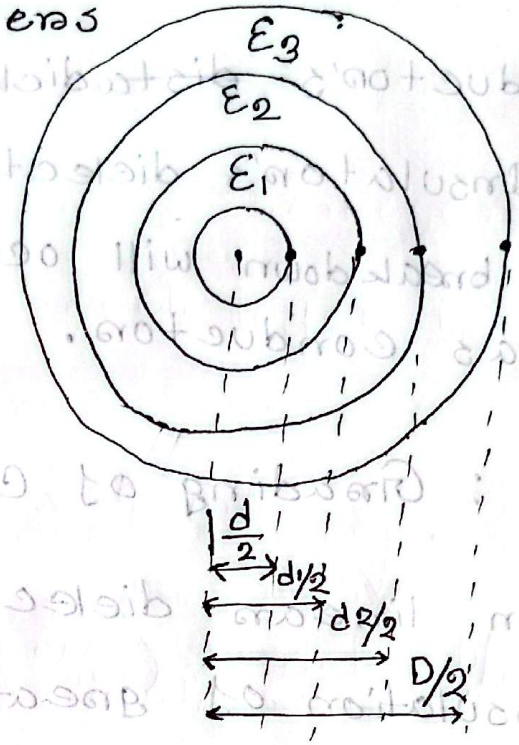
11. Lead to breakdown of insulation.

11.12 : Capacitance grading

Lecture -10

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Art-11.12 Capacitance Grading: The process of achieving uniformity in the dielectric stress by using layers of different dielectric is known as capacitance grading.



$$g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r x}$$

$$g_{1max} = \frac{Q}{\pi\varepsilon_0\varepsilon_1 d}$$

$$g_{2max} = \frac{Q}{\pi\varepsilon_0\varepsilon_2 d}$$

$$g_{3max} = \frac{Q}{\pi\varepsilon_0\varepsilon_3 d}$$

Dielectric stress is equal $g_{1max} = g_{2max} = g_{3max}$.

$$\Rightarrow \frac{1}{\varepsilon_1 d} = \frac{1}{\varepsilon_2 d} = \frac{1}{\varepsilon_3 d}$$

$$\varepsilon_1 d = \varepsilon_2 d = \varepsilon_3 d_2$$

Potential Across first layer,

$$V_1 = \int_{d/2}^{d_1/2} g\,dx = \int_{d/2}^{d_1/2} \frac{Q}{2\pi\varepsilon_0\varepsilon_1 x}\,dx$$

$$= \frac{g_{max} d}{2} \ln\left(\frac{d_1}{d_2}\right)$$

Similarly,

$$V_2 = \frac{g_{max} d_1}{2} \ln\left(\frac{d_2}{d_1}\right)$$

$$V_3 = \frac{g_{max} d_2}{2} \ln\left(\frac{D}{d_2}\right)$$

Total potential difference between core and ~~shield~~ sheath,

$$V = V_1 + V_2 + V_3$$

$$= \frac{g_{max}}{2} \left[d \ln\left(\frac{d_1}{d}\right) + d_1 \ln\left(\frac{d_2}{d_1}\right) + d_2 \ln\left(D/d_2\right) \right]$$

If cable ~~head~~ had homogenous dielectric,

$$V' = \frac{g_{max}}{2} d \ln\left(\frac{D}{d}\right)$$

Here, $V > V'$

- Graded cable can be worked at a greater potential than non graded cable.
- Graded cable's size will be less than non graded cable.

11.13 : Intersheath Grading

$$g_{1max} = \frac{2V_1}{d \ln(d_1/d)}$$

$$g_2 max = \frac{2V_2}{d_1 \ln(d_2/d_1)}$$

$$g_3 max = \frac{2V_3}{d_2 \ln(D/d_2)}$$

since, the dielectric is homogenous.

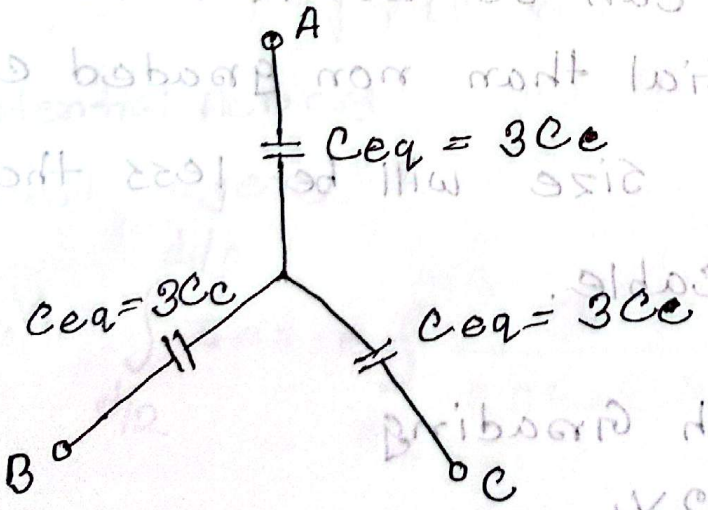
Thus max stress in each layer is the same.

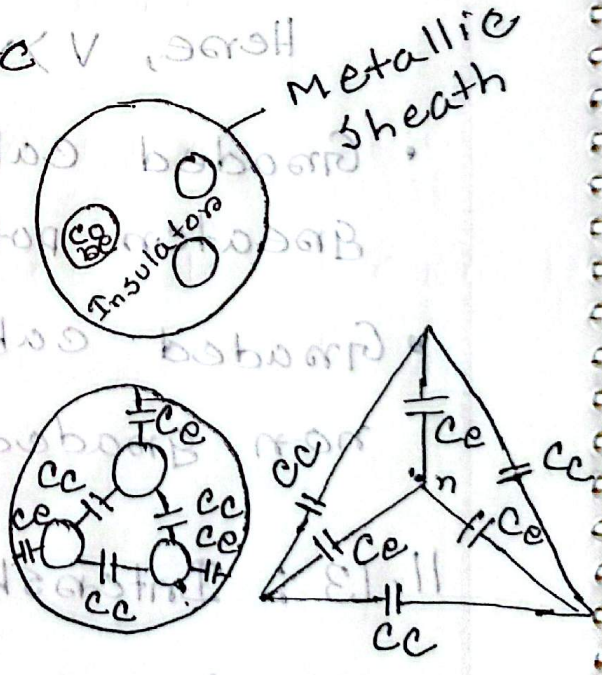
$$g_1 max = g_2 max = g_3 max$$

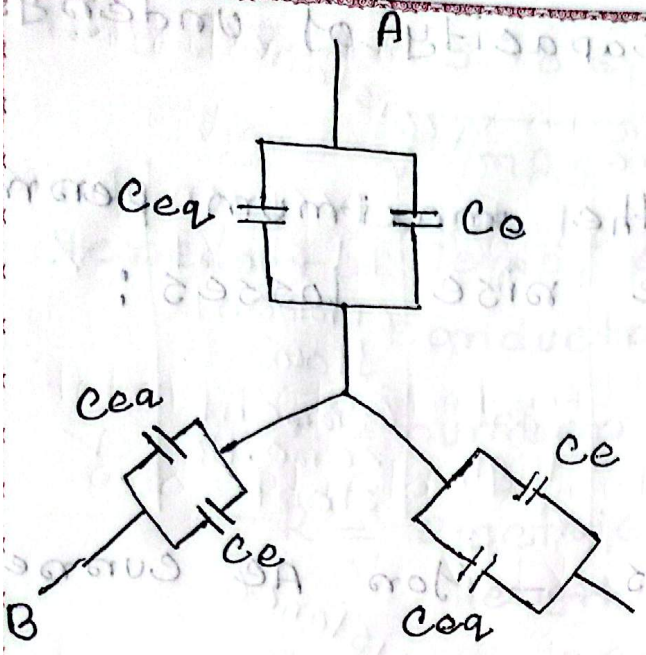
$$\frac{2V_1}{d \ln(d_1/d)} = \frac{2V_2}{d_1 \ln(d_2/d_1)} = \frac{2V_3}{d_2 \ln(D/d_2)}$$

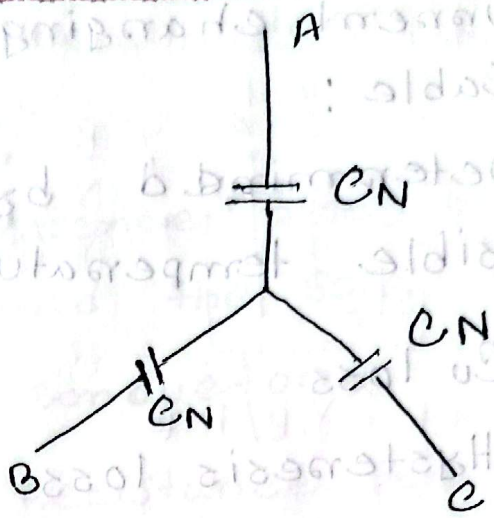
11.14 : Capacitance of a 3 core cable

Δ to Y ; $C_c \rightarrow C_{eq} = 3C_c$









$C_N = C_e + C_{eq}$

$= C_e + 3C_c$

Since, potential difference exists between pairs of conductors and between each conductor and the sheath, electrostatic fields are setup in the cable.

Here, Core conductor capacitance = C_c

Cond. earth cap. = C_e

- Charging current $I_c = \dfrac{V_\phi}{\text{capacitance reactance per phase}}$

$= \dfrac{V_\phi}{X_c}$

$= V_\phi \Big/ \dfrac{1}{2\pi f C_N}$

$= 2\pi f C_N V_\phi$

$= 2\pi f (C_e + 3C_c) V_\phi$

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Current changing Capacity of underground Cable :

- Determined by the maximum permissible temperature rise losses :

1. Cu loss
2. Hysteresis loss
3. Eddy Current loss — for AC current

Safe working Conducton temp. 60°C for armoured cables
50°C for lead sheath cables.

11.17 : Thermal Resistance (S)

$$S = \frac{\text{Temp. difference}}{\text{Heat flowing in a unit time}}$$

$$= \frac{t}{p}$$

Here, k = Thermal resistivity

$$S \propto \frac{l}{a} \quad \therefore S = k\frac{l}{a}$$

$$k = \frac{Sa}{l}$$

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11.19 : Permissible : Current loading

$$I = \sqrt{t/nRS}$$

Here, t = Temp. difference (rise) between conductor and the soil.

n = number of conductors

R = electrical Resistance per metre length

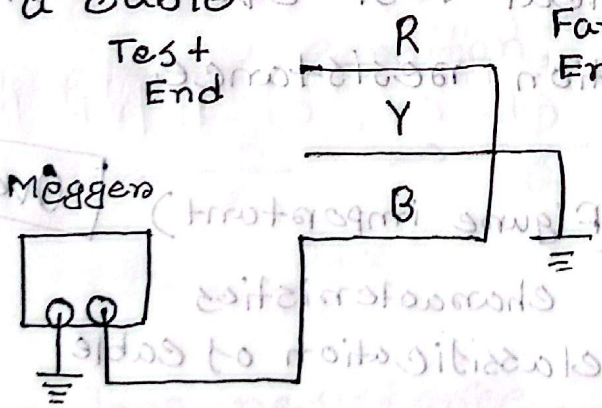
S = Total thermal resistance per metre length.

Art - 11.2.D

Lecture - 11
10.08.26

Types of Cable faults

1. Open circuit fault : Break in the conductor of a cable

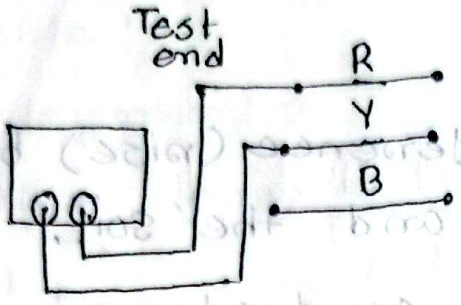


Resistance / Radius	Decision	Indication
0	Current Flow	✓
∞	Current	✗

* If ~~continuous~~ continuity is there, current will flow.

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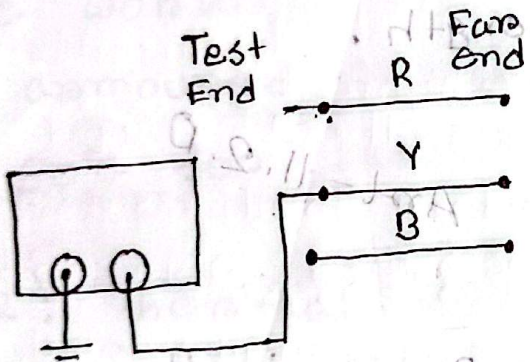
II. Short Circuit fault :



OR, Y.B/R.B

Readings → Resistance	Decision	Indication
0	If short current flow	X
∞ High	Current zero. No short	✓

III. Earth fault :



Readings → Resistance	Decision	Indication
0	Earthing current flow	X
∞	No current will flow	✓

* Megger Electrical test equipment to measure insulation resistance

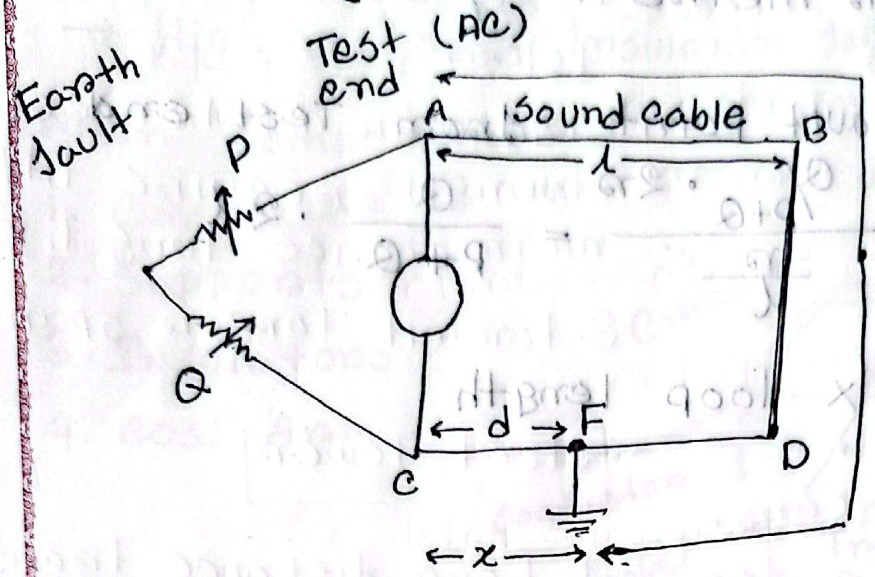
Overview

Chapter 11 → 11.1, (Figure important)
11.3, characteristics
classification of cable
11.6, 11.7, 11.8, 11.9, 11.10, 11.12,
11.13, 11.17, 11.20, 11.21, 11.22

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Art 11.2.2 :

Murray Loop Test :



AB = Sound cable

CD = Faulty cable

* BD এর resistance zero.

R = Resistance of conductor loop upto the fault from the test end.

X = Resistance of other length of the loop.

P, Q, R, X — four ohms of wheatstone Bridge

At Balanced condition,

$$\frac{P}{Q} = \frac{R}{X} \Rightarrow \frac{P}{Q} + 1 = \frac{R}{X} + 1$$

$$\Rightarrow \frac{P+Q}{Q} = \frac{R+X}{X}$$

Here loop resistance, $R + X = 2r$

$$\frac{P+Q}{Q} = \frac{2r}{X}$$

$$\boxed{X = \frac{Q}{P+Q} \cdot 2r}$$

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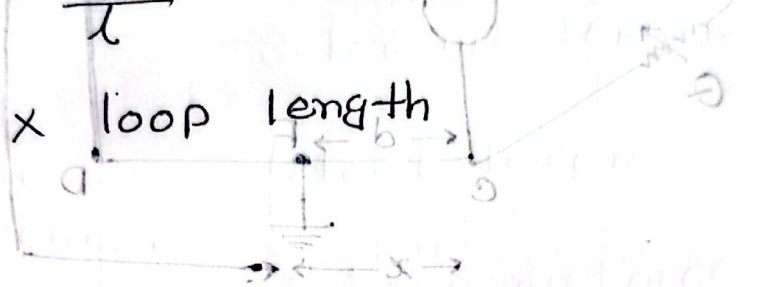
For l metre, resistance is r.

Resistance per metre $= \frac{r}{l}$

Distance of fault point from Test end

$$d = \frac{X}{\frac{r}{L}} = \frac{\frac{Q}{P+Q} \cdot 2r}{\frac{r}{l}} = \frac{Q}{P+Q} \cdot 2l$$

$$d = \frac{Q}{P+Q} \times \text{loop length}$$



Assignment

- Chapter - 08, 11
- Mathematical problem
  Example, Exercise
- Derivations
- Submission : 07.09.2026

TT : 07.09.2026

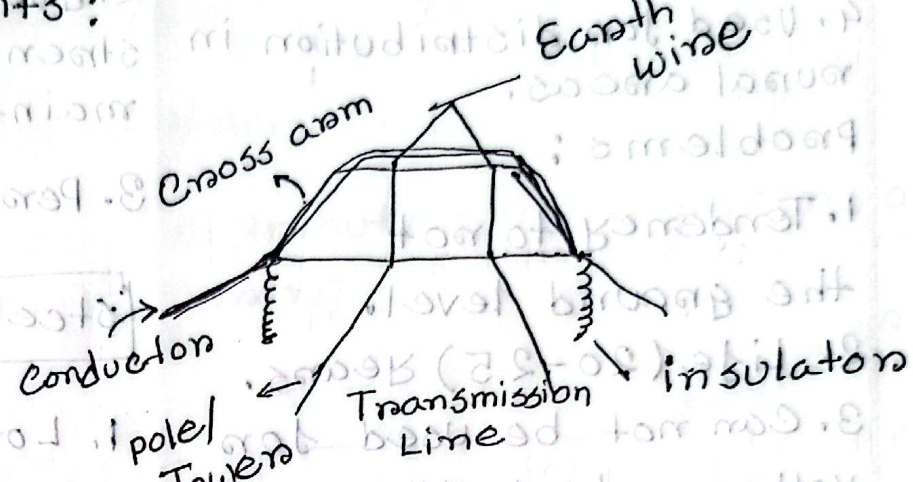
↳ Syllabus : Ch- 08, Ch-11 [V.k Mehta]

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Chapter-08: Mechanical Design of Overhead Wire

→ Installation cost
→ High Voltage (insulation failure) } Underground Cable

Main Components:
1. Conductor
2. Supports
3. Insulators
4. Cost Arm



conductor Materials:
i. High electrical conductivity
ii. High strength
iii. Low cost
Example: Copper, Al, steel cored Al.

Art 8.3: Line support (pole/ Tower)
1. High mechanical strength
2. Light in weight
3. cheap in cost
4. Longer life

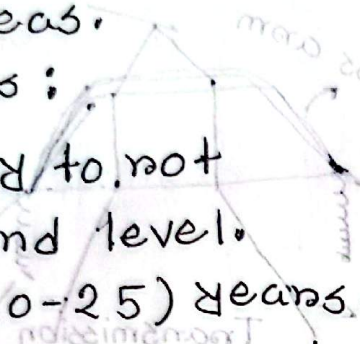
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Wooden poles

1. Made of woods
2. Span (50 metre)
3. Cheap, easily available
4. Used for distribution in rural areas.

Problems :

1. Tendency to rot the ground level.
2. Life (20-25) years.
3. Can not be used for voltages high than 20kV.



RCC poles

1. Reinforce concrete poles
2. Greater mechanical strength, require little maintanance.
3. Transport cost High.

~~Problems of wooden poles :~~

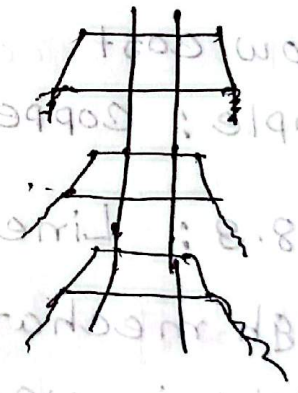
~~1. Tende~~

Steel poles

1. Substitute of wooden poles.
2. Greater mechanical strength, require little maintanance. long life.
3. Permit larger span.

Steel Tower

1. Long distance trans mission, at high voltage.
2. Larger spans.



Steel Tower

- Long distance transmission at highers voltage
- longer spans

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Ans 8.5 : Types of insulators

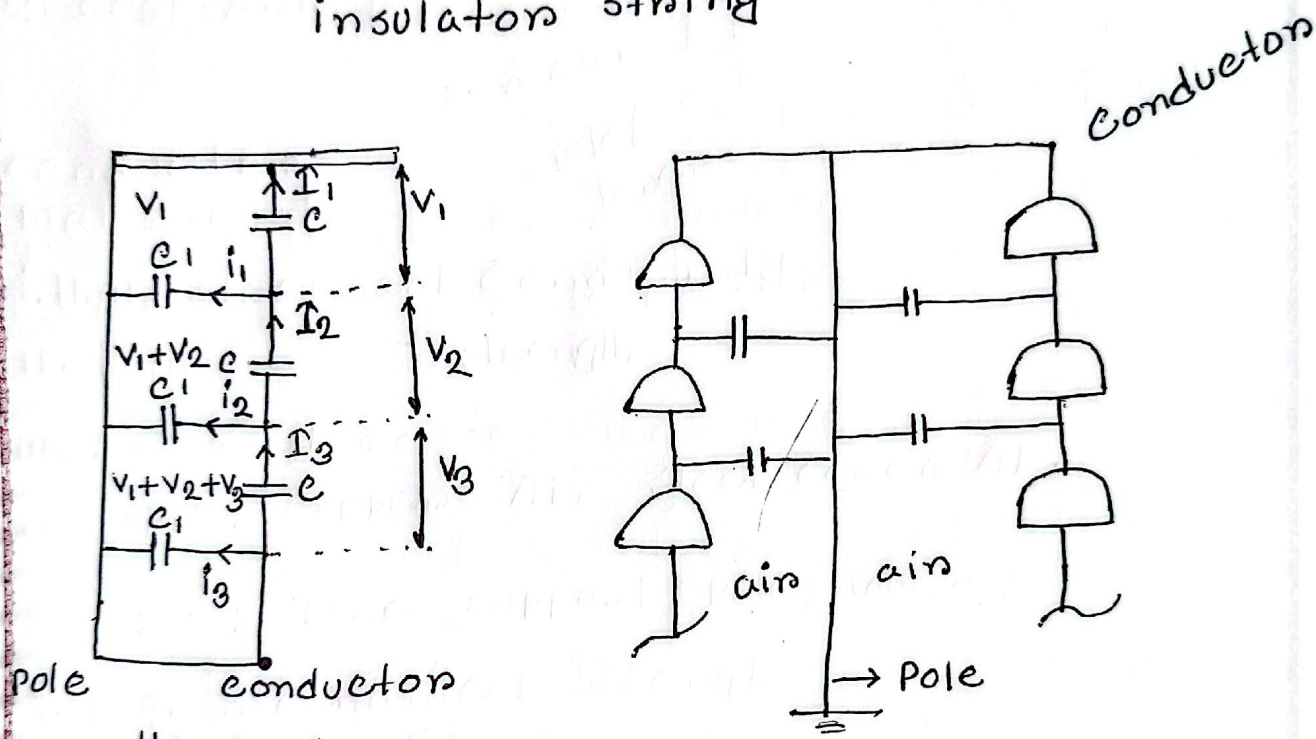
** 1. Pin Type < 33 kV

** 2. Suspension Type > 33kV

3. Strain insulator

4. Shackle insulator

Ans 8.6 : Potential Distribution over suspension insulator string



Here, C_1 = Shunt capac.

C = Shell capacitance

Pole, conductor → behaves like Conductor

air, insulation disc → behaves like insulator

So, Capacitance will form between them

চটা conductor এর সাথে ∆, capacitor এর মতো কাজ করবে।

Ans 8.7

$$\text{String Efficiency} = \frac{\text{voltage across the string}}{n \times \text{voltage across disc nearest to conductor.}}$$

Here, n = number of discs in the string.

The ratio of voltage across the whole string to the product of number of discs and the voltage across the disc nearest to the conductor is known as string efficiency.